

CBSE Class 10 Mathematics — Quadratic Equations (Hard)

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains questions.Allquestionsarecompulsory.
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. The discriminant of the quadratic equation $x^2 - 5x + 6 = 0$ is:

- (A) (A) 1
- (B) (B) -1
- (C) (C) 49
- (D) (D) 0

(simple)

2. The discriminant of the quadratic equation $2x^2 - 3x + 1 = 0$ is:

- (A) (A) 1
- (B) (B) -1
- (C) (C) 9
- (D) (D) -7

(simple)

3. The discriminant of the quadratic equation $2x^2 - 3x + 5 = 0$ is:

- (A) (A) 31
- (B) (B) -31
- (C) (C) 41
- (D) (D) -41

(simple)

4. The roots of the quadratic equation $x^2 - 7x + 10 = 0$ are:

- (A) (A) 2, 5
- (B) (B) -2, -5
- (C) (C) 2, -5

(D) (D) -2, 5

(simple)

5. Which of the following is the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$?

(A) (A) $\sqrt{40}$

(B) (B) 4

(C) (C) $16 - 24$

(D) (D) 0

(simple)

6. Assertion (A): The quadratic equation $2x^2 - 5x + 3 = 0$ has two distinct real roots.
Reason (R): For a quadratic equation $ax^2 + bx + c = 0$, if $b^2 - 4ac > 0$, then the equation has two distinct real roots.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

(simple)

7. Assertion (A): The quadratic equation $x^2 - 3x + 5 = 0$ has no real roots. Reason (R): For a quadratic equation $ax^2 + bx + c = 0$, if discriminant $D = b^2 - 4ac < 0$, then the equation has no real roots.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

(simple)

Section B: Very Short Answer (2 marks each)

1. Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$. *(simple)*

2. Find the discriminant of the quadratic equation $2x^2 - 3x + 1 = 0$. *(simple)*

3. Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$. *(simple)*

Section C: Short Answer I (3 marks each)

1. Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$. Hence, determine the nature of its roots. *(simple)*
2. Find the nature of the roots of the quadratic equation $2x^2 - 4x + 3 = 0$. *(simple)*
3. Find the nature of the roots of the quadratic equation $2x^2 - 4x + 3 = 0$. *(simple)*
4. Find the discriminant of the quadratic equation $3x^2 - 5x + 2 = 0$ and hence determine the nature of its roots. *(simple)*
5. Find the discriminant of the quadratic equation $3x^2 - 6x + 2 = 0$ and hence determine the nature of its roots. *(simple)*

Section D: Long Answer (4-5 marks each)

1. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train. *(medium)*
2. The speed of a boat in still water is 8 km/h. It takes the same time to travel 15 km upstream as it takes to travel 25 km downstream. Find the speed of the stream. *(medium)*
3. The sum of the squares of two consecutive positive integers is 365. Find the integers. *(medium)*
4. The sum of two numbers is 15. If the sum of their reciprocals is $\frac{3}{10}$, find the numbers. *(medium)*
5. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train. *(medium)*
6. A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. Find the original speed of the train. Also, find the time taken by the train to cover the distance at the original speed. *(hard)*
7. A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. Find the original speed of the train. Also, find the time taken at the original speed. *(hard)*
8. A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. Find the original speed of the train. Also, find the time taken at the original speed. *(hard)*
9. A motorboat whose speed in still water is 18 km/h takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream. *(hard)*

10. A motorboat whose speed in still water is 18 km/h takes 1 hour more to go 48 km upstream than to return to the same point downstream. Find the speed of the stream.
(hard)